

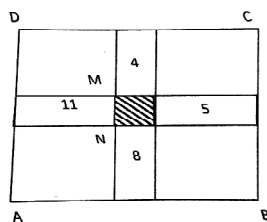
AMTI (NMTC) - 2012

RAMANUJAN CONTEST - FINAL - PRIMARY LEVEL

- Find the sum (S) of all numbers with 2012 digits and digital sum 2. Find also the digital sum of S.
- A number 'n' is called a "lonely odd composite" number if
 - n is an odd composite number and
 - both (n - 2) and (n + 2) are prime numbers.
- In the adjoining figure ABCD is a parallelogram of perimeter 21.

It is subdivided into smaller parallelograms by drawing lines parallel to the sides.

The numbers shown are the respective perimeters of the parallelograms in which they are marked. (For example the perimeter of the parallelogram $\angle MNP$ is 11). Find the perimeter of the shaded parallelogram.



- l and b are two numbers of the form $\frac{p}{q}$ where p and q are natural numbers. Further l, b are greater than 2.
- a, b, c, d are the units digits of four natural numbers each of which has four digits. The tens digit of these four numbers are the 9 complements of the units digit. The hundreds digits are the 18 complements of the sum of their respective tens and units digits. The thousands digits are the 27 complements of the sum of their respective hundreds, tens and units digits. If $a + b + c + d = 10$, find the sum of these four numbers. {9 complement of a number $4x$ is $9 - x$, 18 complement of a number y is $18 - y$, 27 complement of a number z is $27 - x$ }.
- A sequence is generated starting with the first term t_1 as a four digit natural number. The second third and fourth terms (t_2 , t_3 and t_4) are got by squaring the sum of the digits of the preceding terms. (Ex. $t_1 = 9999$ then $t_2 = (9 + 9 + 9 + 9)^2 = 36^2 = 1296$, $t_3 = (1 + 2 + 9 + 6)^2 = 32^2 = 1024$, $t_4 = (3 + 2 + 4)^2 = 81$. Start with $t_1 = 2012$. Form the sequence and find the sum of the first 2012 terms.
- Find the two digit numbers that are divisible by the sum of their digits. Give detailed solution with logical arguments.
- ABCD is a square and the sides are extended as shown in the diagram. The exterior angles are bisected and the bisectors extended to form a quadrilateral PQRS. Prove that PQRS is a square.

